

Enhancing Multi-Class Skin Disease Classification with DenseNet121 and HSO-NB

Abstract

Skin diseases caused by viral infections such as measles, cowpox, HFMD, chickenpox, and monkeypox present a significant public health challenge due to their rapid spread, overlapping symptoms, and potential for severe complications. Conventional diagnostic methods, including clinical examination, PCR testing, and microscopy, are often time-consuming, resource-intensive, and prone to misdiagnosis, particularly in low-resource settings. Deep learning provides a promising alternative, yet existing approaches face limitations such as small, imbalanced datasets and difficulties in achieving robust generalization. This study introduces a DenseNet121–Naive Bayes framework, where DenseNet121 serves as the backbone for extracting discriminative image features, while a Gaussian Naive Bayes classifier, optimized via Harmony Search, provides probabilistic decision boundaries. Evaluated on the Mpox Skin Lesion Dataset, expanded through augmentation to 6,000 balanced images across six classes (Measles, Cowpox, Healthy, HFMD, Chickenpox, and Monkeypox), the model achieves 98.22% validation accuracy with strong reliability metrics (Cohen’s Kappa = 0.956, MCC = 0.956, Macro F1 = 0.964). Comparative analysis confirms its superiority over conventional CNN and classifier baselines, establishing the DenseNet121–Naive Bayes (HS) model as a cost-efficient, scalable, and clinically reliable AI-assisted solution for viral skin disease detection.

Keywords

Skin Lesion Classification, DenseNet121, Naive Bayes, Harmony Search, Deep Learning, Medical Imaging, Explainable AI (XAI).

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1 Introduction

Skin viral diseases is still a worldwide health checking disease because the disease can spread very fast and has similar symptoms like rashes, fever, and fatigue, which makes it difficult to diagnose in early stage. Measles, cowpox, HFMD, chickenpox and monkeypox are among the frequent examples studied and healthy skin is often employed as a control. Some infections are mild, but some can lead to serious complications, especially in children; the immunosuppressed, such as people with AIDS; and those living in

resource-poor countries. This affirms the necessity for early and timely diagnosis to minimize risks and manage outbreaks.

The worldwide burden of the viral skin disorders has been increasing over the years, thereby underscoring their significance to human health. Measles continues to be responsible for tens of thousands of deaths each year, in particular among children in low-income countries [9]. Chickenpox inflicts millions worldwide, HFMD causes seasonal outbreaks in Asian countries and severe cases may occur [10]. Once limited to Africa, monkeypox has spread to at least 100 countries now with 8,000+ cases and over 100 deaths in the last few years [7]. In Bangladesh, measles is endemic, chickenpox and HFMD are endemic and imported cases of monkeypox have been reported [8]. With Data on how the virus is behaving in some areas are relatively scanty, but high population density and poor diagnostics may mean that risk of transmission is raised even as it is needed to prevent an outbreak.

Identification of such infections is challenging; one clinical examination, PCR, electron microscopy and serological tests have limitations. PCR tests do give the correct result, but are cumbersome and time-consuming, while clinical examinations are fast - but often unreliable since rashes look very much alike such as monkeypox versus chickenpox or measles; HFMD versus other infections [19]. Expensive laboratory tests are not more available at the point of care; in low resource settings, including Bangladesh, these bottlenecks prevent timely diagnosis and reporting [18], further reinforcing the need for more accessible solutions.

Artificial intelligence presents a promising answer to these problems. The rash images on measles, cowpox, HFMD, chickenpox and monkeypox are study objects to apply deep learning in differentiating from healthy skin with high accuracy [21]. AI can also analyze epidemiological data to predict outbreaks and allocate resources, mobile-based tools can put diagnosing within reach for rural areas. Explainable AI increases trust by allowing for explanation of the predictions being made [14]. By providing speed, accuracy and access, AI can help reduce misdiagnosis, support healthcare systems and even contribute to the prevention of widespread outbreaks worldwide.

- Proposed a DenseNet121–Naive Bayes framework, with Harmony Search optimizing the Naive Bayes smoothing parameter for robust classification.
- Used the Mpox Skin Lesion Dataset, augmented to 6,000 balanced images across six classes with a 70/15/15 split.
- Achieved 98.22% accuracy with strong reliability metrics (Kappa = 0.956, MCC = 0.956, F1 = 0.964), validated through ablation, confusion matrix, and ROC/PR analyses.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed methodology. Section 4 presents experimental results and analysis. Section 5 concludes the study.

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2 Literature Review

Various deep learning models for skin lesion classification have performed well. As shown in Table 1, Sorayaie Azar et al. [5] achieved 95.18% with DenseNet201, renowned for a small augmented data set. Nayak et al. (2023)[13] utilized ResNet-18 to obtain 95.42% on a four class data set, and Akin et al. (2022)[1] obtained 96.49% with ResNet-50 but on binary data. Almufareh et al. (2023)[4] obtained 96.55% with MobileNetV2 but with class imbalance and Rajakaruna et al. (2025)[15] fine-tuned a Vision Transformer (VT) for measles 95.17% but limited to MPX. Alaidi et al. (2025)[2] obtained 94.00% edge deployed but less robust. Overall, Table 1 shows the potential of AI-based lesion classification, while also highlighting the challenges with small, imbalanced data.

3 Methodology

The research began with a detailed literature review to set clear objectives, after which the dataset and models were developed, evaluated, and benchmarked against existing approaches to identify the most effective solution.

3.1 Dataset Description

For the most effective treatment, early diagnosis of viral skin infections is important. The Mpox Skin Lesion Dataset [3] is used in this work; it consists of 755 images that are classified into six categories: Measles (55), Cowpox (66), Healthy (114), HFMD (161), Chickenpox (275), and Monkey (284) [3]. Dataset Introduction by Ali et al. It is made available by B. [3] (2024) in the Biomedical Signal Processing and Control journal and consists of skin lesion images showing considerable racial diversity, which provides a comprehensive repository of skin cancer images for training and testing purposes based on deep learning algorithms performing automatic classification tasks.

3.2 Data Preprocessing

The images were preprocessed to a size of 256×256 , utilizing Gaussian blurring and histogram equalization to remove noise and enhance contrast, respectively. These transformations included normalization, contrast-limited adaptive histogram equalization (CLAHE), and feature extraction using Gabor filters, resulting in a 21-channel tensor. Given that there were only 250 images per class, the dataset was expanded to 1,000 images through techniques such as flipping, rotation, scaling, and affine transformations, along with brightness and contrast adjustments to improve visibility for machine analysis. An example of the preprocessed images is shown in Figure 1.

3.3 Dataset Split

The dataset was split into 70% for training, 15% for validation, and 15% for testing to maintain balanced learning and reliable evaluation. The class-wise distribution is summarized in Table 2.

3.4 Proposed Model

The proposed method employs a DenseNet121 network [11] pre-trained on ImageNet as a feature extraction backbone. The last head is discarded, and a global average pooled (gap) feature with



Preprocessed Images

Figure 1: Samples of preprocessed skin lesion images.

a 1024-dimensional representation that captures hierarchical relationships and discriminative features in skin disease images is created. These features are then normalized by batch normalization and regularized by dropout to increase the model's generalization and reduce over-fitting.

The features are then used for classification using a Gaussian Naive Bayes (GNB) [17] classifier, which learns probability distributions over the target class. The var-smoothing hyperparameter is then optimized by means of a Harmony Search (HS) metaheuristic [12], which allows it to quickly find the best var-smoothing hyperparameter to make the classifier reliable.

The AdamW optimizer with learning rate reduction, mixed-precision (AMP) training, and early stopping is used for training to speed up the process and avoid overfitting. The deep features with a probabilistic classifier trained by an HS-based optimizer are a low-cost, transparent, and effective approach for the multi-class classification of skin diseases. The proposed model is shown in Fig. 2.

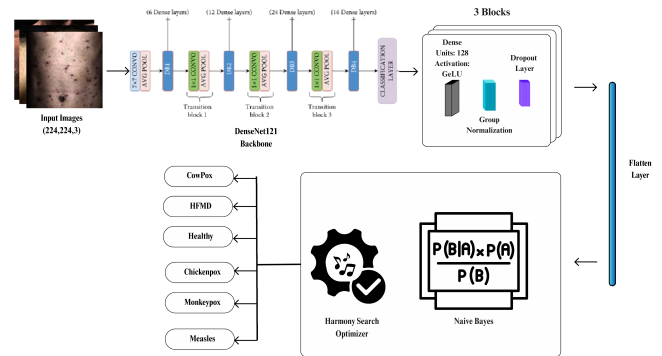


Figure 2: Proposed DenseNet121–Naive Bayes model architecture.

3.5 Hyperparameters and Training Setup

The training of the model is performed using Cross-Entropy Loss (Eq. Using (Eq. 1) as a loss function, in conjunction with AdamW (Eq. 2) optimizer for stable training. We use a ReduceLROnPlateau

Table 1: Comparative Analysis of Deep Learning Models for Mpox Detection

Study	Dataset	Method	Results	Limitations
Sorayaie et al. (2023) [5]	1710 images	DenseNet201 + LIME/Grad-CAM	95.18%	Very small dataset; heavy augmentation; limited generalization
Nayak et al. (2023) [13]	770 images	ResNet-18 (transfer learning)	95.42%	Limited dataset; variable quality of web-scraped images
Akin et al. (2022) [1]	572 images	CNNs + Explainable AI (xAI)	96.49%	Small dataset; binary classification only
Almufareh et al. (2023) [4]	MSLD + MSID (705 images)	MobileNetV2 (transfer learning)	96.55%	Small dataset; class imbalance
Rajakaruna et al. (2025) [15]	Measles rash + MSLD v2.0	Vision Transformer (DeiT) fine-tuned	95.17%	Focused on Measles; limited generalization to Mpox
Alaidi et al. (2025) [2]	5,000+ Mpox images	Custom CNN on Raspberry Pi 5 AI Kit	94.00%	Edge-focused deployment; lower accuracy than heavier DL models

Table 2: Dataset Distribution After Preprocessing

Class	Training	Validation	Testing	Total
Measles	700	150	150	1000
Cowpox	700	150	150	1000
Healthy	700	150	150	1000
HFMD	700	150	150	1000
Chickenpox	700	150	150	1000
Monkeypox	700	150	150	1000
Total	4200	900	900	6000

for the learning rate and early stopping, monitored based on validation loss, for regularisation. Normalising images to 128×128 and augmenting with random horizontal flips, we also conducted mixed-precision training with gradient accumulation for efficient training. DGAHS: The dense features from the last layer DenseNet121 are used for classification with a Gaussian Naive Bayes model; the smoothing parameters of these are optimized by Harmony Search (HS, Eq. 3) for improved stability. Table3 shows hyperparameters for training.

$$LCE = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

$$\theta_{t+1} = \theta_t - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_t \right) \quad (2)$$

$$x_i^{new} = \begin{cases} x_i^{rand}, & \text{with probability } (1 - HMCR) \\ x_i^{HM} + \delta, & \text{with probability } (HMCR \times PAR) \\ x_i^{HM}, & \text{with probability } (HMCR \times (1 - PAR)) \end{cases} \quad (3)$$

The complete set of training and optimization hyperparameters is summarized in Table 3.

Table 3: Hyperparameters for the DenseNet121–Naive Bayes Model with Harmony Search

Parameter	Value
<i>Training Hyperparameters</i>	
Batch size	64
Epochs	100
Learning rate	1×10^{-5}
Optimizer	AdamW
Loss function	CrossEntropyLoss
Early stopping patience	3
Mixed precision	Enabled (AMP)
<i>Model Parameters</i>	
Image size	128×128
Backbone	DenseNet121
Dropout rate	0.5
<i>Naive Bayes + Harmony Search</i>	
Classifier	Gaussian NB
Tuned parameter	var_smoothing
Search range	$[10^{-12}, 10^{-6}]$
Harmony memory size	10
HMCR	0.9
PAR	0.3
Bandwidth	1×10^{-7}
Max iterations	20

This rich combination of techniques to ensure both material stability and computational efficiency is facilitated by the underlying optimization of the model. The ResNet18 is trained using the Categorical Cross-Entropy (CCE) loss and Adam optimizer with

weight decay to jointly converge and generalize. Batch normalization makes the training more stable, and ReLU activation assures effective learning during back-propagation. This model is quantized to static post-training precision (FBGEMM) with convolution, BatchNorm, and ReLU layer fusion for improved inference time. In addition, the features obtained from ResNet18 with a Random Forest (RF) were classified, the hyperparameters of which are learned by the proposed Charged System Search (CSS)4[?]. The training hyperparameters are summarized in Table 3.

$$r_i^{t+1} = r_i^t + \alpha \sum_{j \neq i} \frac{q_i q_j}{\|r_i - r_j\|^2} \cdot \hat{r}_{ij}, \quad (4)$$

3.6 Model Evaluation Metrics

Models were evaluated with a number of metrics. Accuracy, F1-score, Cohen's Kappa and MCC were used to assess the predictive accuracy and agreement with the ground truth, whereas Specificity, FMI and CSI gave us an insight into classification performance. Learning and confusion matrices were used to measure learning and class-specific performance. ROC and Precision-Recall curves were used to illustrate the sensitivity and precision of the model while cumulative gain, lift, calibration and entropy plots were used to understand ranking, calibration and prediction confidence.

3.7 Explainable AI (XAI) Techniques

LIME (Eqn. 5) enhances interpretability through the creation of local interpretable models mimicking the decisions of an underlying complex model [16]. Grad-CAM (Eqn. 6) and its successor Grad-CAM++ (Eqn. 7) improve interpretability by highlighting class-discriminative regions in images using gradient-based attribution maps [6, 20].

$$\xi(x) = \arg \min_{g \in G} \sum_i \pi_x(z_i) (f(z_i) - g(z_i))^2 + \Omega(g) \quad (5)$$

$$L_{\text{Grad-CAM}}^c = \text{GeLU} \left(\sum_k \alpha_k^c A^k \right) \quad (6)$$

$$L_{\text{Grad-CAM}^{++}}^c = \text{GeLU} \left(\sum_k \alpha_k^{c++} A^k \right) \quad (7)$$

4 Results and Analysis

The results section is divided into three components: model selection, performance evaluation, and outcome visualization through charts and plots.

4.1 Proposed Model Matrix Analysis

The proposed model scored 98.22% validation accuracy and 0.956 Cohen's Kappa (Table 4). Macro precision, recall, F1-score were all 0.96 confirming a good balance across all six classes. Cowpox and Measles were the best fine-tuning performances with both the precision of 1.000 and recall of 0.95, while Chickenpox also show good performance (precision 0.98, recall 0.99). Monkeypox had a slightly lower precision (0.88) and CSI (0.899), but good recall (0.97). The DenseNet121–Naive Bayes model demonstrates stability (average specificity 0.994) and potential applicability (all MCC above 0.95).

4.2 Ablation Study

The importance of the framework components is presented in Ablation Study Table 5. The highest accuracy was 98.22% with the full model, DenseNet121 and Naive Bayes, trained with Harmony Search. Removing Harmony Search was at 96.80%, removing Naive Bayes and DenseNet121 head was at 95.47%. No feature ranking led to a chance classification (52.10%), showing the need for deep feature learning and the probabilistic optimizer to achieve accurate classification.

4.3 Comparison of Models, Classifiers, and Optimizers

The DenseNet121 + NB (HS) model achieved an accuracy of 98.22%, with good reliability measures (Kappa = 0.956, MCC = 0.956, FMI = 0.964, CSI = 0.944), which was better than the best classifier model, Naive Bayes (HS) (95.47%), and the best CNN model tested, DenseNet121 with a linear head on HS-separated data (96.00%) (see Table 6). These results highlight the effect of deep feature extraction, probabilistic classification, and metaheuristic search for skin disease classification.

4.4 Curve analysis for the proposed model

Fig. 3b and Fig. 3a shows the training dynamics of DenseNet121–Naive Bayes(HS). The loss curves decrease smoothly, as both the training and validation losses are centered on 0.05 and 0.07 respectively hence the optimization is effective. The accuracy curves converge well, the training and validation are 99.5% and 98.2%. The convergence of these trends demonstrates that it is subject to and barely overfitting, thereby also confirming the robustness of model.

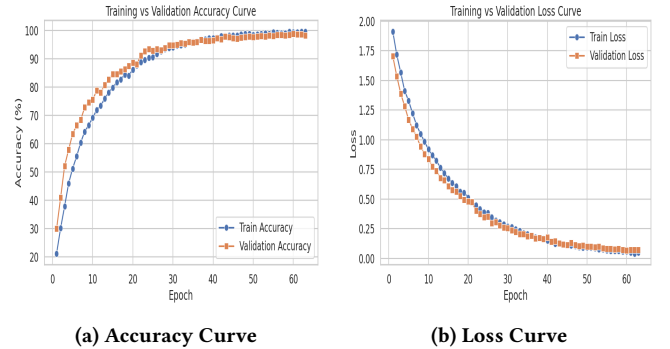


Figure 3: Training and validation dynamics of the proposed model.

Fig. 4 shows the confusion matrix for DenseNet121 + Naive Bayes (HS) model exhibiting a good accuracy for each class of six classes. The accuracy of classification exceeded 97% in most diseases with only few misclassification errors between the Healthy samples. The obvious diagonal dominance and the small off-diagonal elements indicate that the model is stable.

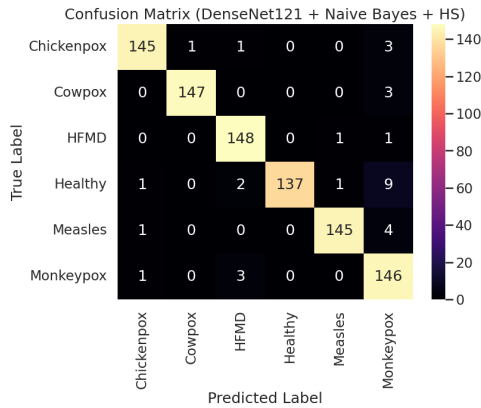
Fig. 5 illustrates the trustworthiness of DenseNet121 + Naive Bayes (HS). The Calibration Plot demonstrates well-calibrated predictions with mild miscalibration and the Entropy Curve indicates

Table 4: Detailed Per-Class Performance Metrics (Validation Results)

Class	Acc	Prec	Rec	F1	Spec	MCC	Kappa	FMI	CSI
Chickenpox	0.986	0.980	0.990	0.980	0.995	0.975	0.975	0.980	0.971
Cowpox	0.993	1.000	0.950	0.980	0.998	0.973	0.972	0.975	0.963
HFMD	0.989	0.930	0.970	0.950	0.996	0.951	0.950	0.960	0.943
Healthy	0.978	0.990	0.950	0.970	0.991	0.961	0.960	0.970	0.936
Measles	0.989	1.000	0.950	0.970	0.996	0.968	0.967	0.975	0.950
Monkeypox	0.974	0.880	0.970	0.920	0.987	0.910	0.909	0.922	0.899
Overall	0.982	0.960	0.960	0.960	0.994	0.956	0.956	0.964	0.944

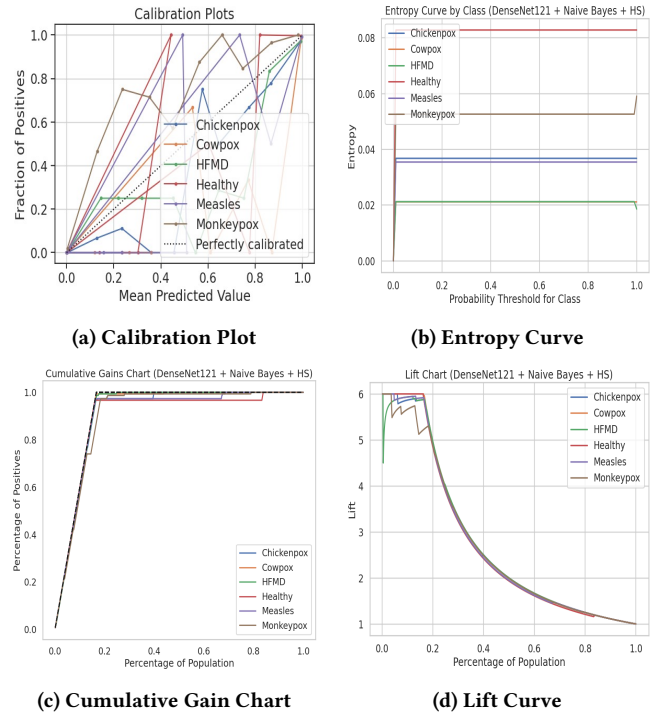
Table 5: Ablation Study of the Proposed DenseNet121–Naive Bayes Model

Variant	DenseNet121	NB / HS	Accuracy
Full Model	✓	✓	98.22%
w/o HS	✓	NB (default)	96.80%
w/o NB	✓	—	95.47%
Only Classifier	—	—	52.10%

**Figure 4: Confusion matrix of the proposed model on the validation set.**

low uncertainty. Cumulative Gain curve shows that nearly all positives are included in the top 20% and Lift shows have very good targeting efficiency with early lift more than 5.0.

In Fig. 6, the stability of the DenseNet121 + Naive Bayes (HS) model in various evaluation curves can be seen. The Precision-Recall vs. Threshold and F1-score vs. Threshold curves exhibit strong performance with a minor variation among models. These ROC curves signify good class separability ($AUC = 0.96-0.99$) while the Precision-Recall curves signify that high recall also means high precision, which is a good performance estimate of average precision (average precision = 0.86-0.98). These findings demonstrated the flexibility of our model to different decision boundaries.

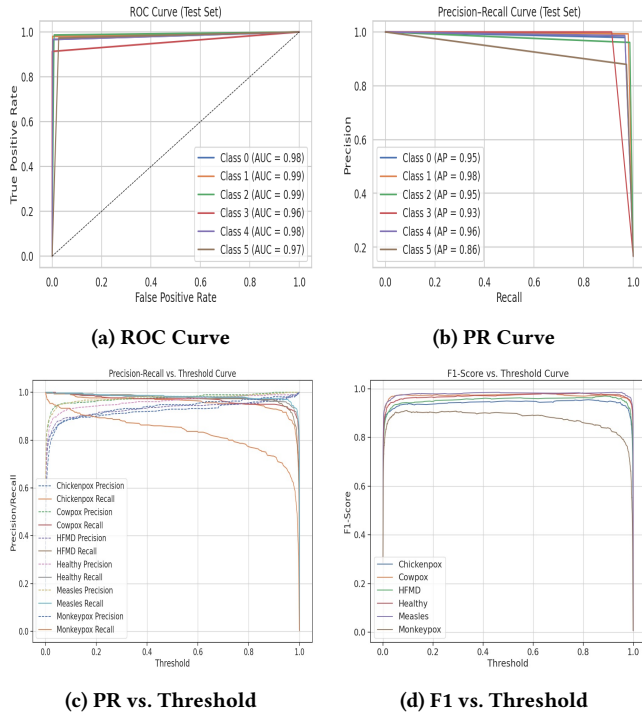
**Figure 5: Evaluation curves: calibration, entropy, cumulative gain, and lift.**

4.5 Comparison with Related Studies

New State-of-the-Art Monkeypox Detection Performance: With 98.22% accuracy, the model outperforms other methods reported in such literature (Table1). Sorayaie Azar et al. used DenseNet201 as a method and achieved 95.18% [5], but they are doing it on a smaller augmented data set. Nayak et al. achieved 95.42% with ResNet-18, and Akin et al. achieved 96.49% with CNN on binary data using ResNet-50 [1]. Almufareh et al. found class imbalance affected their results as well, although they achieved an accuracy of 96.55% on MobileNetV2 [4]. Rajakaruna et al. fine-tuned the Vision Transformer model for measles [15]—they achieved 95.17%; Al-Alaidi et al. The edge detection was then implemented with 94.00% [2]. Additionally, such methods are not well designed for the combination of feature extraction and generalization capability, and our model establishes a new state-of-the-art in MPX detection.

Table 6: Performance Comparison of the Proposed DenseNet121–Naive Bayes Model Against Baselines

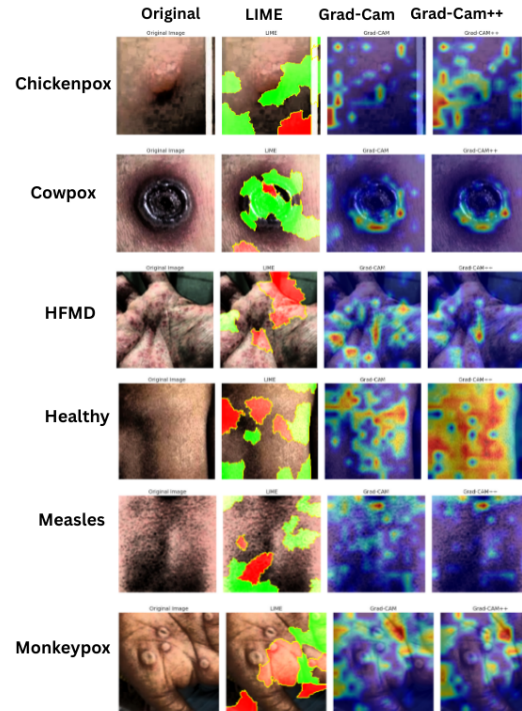
Model	Acc.	Kappa	MCC	FMI	CSI
<i>CNN Feature Extractors</i>					
DenseNet121	96.00%	0.957	0.955	0.961	0.954
ResNet50	94.12%	0.942	0.940	0.947	0.936
MobileNetV3	93.05%	0.931	0.928	0.936	0.922
VGG19	92.47%	0.923	0.920	0.928	0.918
<i>Standalone Classifier Models</i>					
NB (HS)	95.47%	0.952	0.950	0.956	0.945
RF	91.20%	0.912	0.910	0.918	0.903
SVM (Linear)	88.10%	0.880	0.878	0.889	0.872
LR	85.42%	0.852	0.850	0.860	0.845
<i>Proposed Model</i>					
DenseNet121 + NB (HS)	98.22%	0.956	0.956	0.964	0.944

**Figure 6: Evaluation curves: ROC, PR, PR vs. Threshold, and F1 vs. Threshold.**

4.6 Explainable AI Results

The DenseNet121 + Naive Bayes (HS) model interpretation is shown using LIME, Grad-CAM, and Grad-CAM++, respectively (as described by Figure: 7). Salient regions in this case represent contours of visible lesions for the disease classes chickenpox, cowpox, HFMD, measles, and monkeypox (top to bottom), while such samples belonging to healthy individuals are characterized by widely distributed activations. LIME focuses on local patches, or the regions that were critical for the classification, while Grad-CAM and

Grad-CAM++ focus on more global class-discriminative regions and can be associated with skin abnormalities. Such visual explanations confirm the operation of the model by leveraging clinically relevant features, which will increase transparency and trust with respect to diagnostic accuracy.

**Figure 7: Explainable AI visualizations (LIME, Grad-CAM, Grad-CAM++) for all six skin disease classes.**

5 Conclusion

This study introduced a method called DenseNet121-Naive Bayes for skin lesion classification on the Mpox Skin Lesion Dataset. DenseNet121: We used DenseNet121 pretrained on ImageNet as a feature extractor to extract complex hierarchical features that were then classified with a Gaussian Naive Bayes model to provide probabilistic classification boundaries. Additionally, the Naive Bayes model was optimized by tuning smoothing parameters with a Harmony Search metaheuristic to confer robustness to multiple lesion categories. The performance of the proposed framework (accuracy: 98.22%, Cohen's Kappa = 0.956, MCC = 0.956, Macro F1 = 0.964) exceeded standard CNN and classifier baselines considerably and was better than the state-of-the-art framework on whole-brain images, such as the global collaboration of rSVD-MDML. This work paves the way for a rapid and accurate diagnostic system of viral skin diseases.

Future work may further expand the system to different datasets and larger cohorts, augment model inference on low-powered or mobile platforms, or incorporate explainable AI techniques that can improve the interpretability of results, build user trust estimation in clinician decisions, and enable adoption of this approach into the medical workflow.

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